

Labwork 6: Mapping

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1 Implementation

Three CUDA image-processing operations were implemented using a 2D grid and 2D blocks. For binarization, each grayscale pixel is compared with 128 and converted to either 0 or 255. Brightness control increases the value of each RGB channel. Image blending combines two images using

$$Q(x, y) = c\Phi_1(x, y) + (1 - c)\Phi_2(x, y).$$

Each CUDA thread processes one pixel using its (x, y) coordinates.

2 2D Block Size Experiment

Different 2D block sizes were tested: $(8, 8)$, $(16, 16)$, $(32, 16)$, $(16, 32)$, and $(32, 32)$. For each configuration, the execution time was measured five times and the average value was calculated.

For binarization, the fastest average time was approximately 0.000139 s with $(32, 16)$. For brightness control, the best result was obtained with $(16, 32)$ at approximately 0.000159 s.

For image blending, $(8, 8)$ achieved the fastest average execution time of approximately 0.000262 s. The other block sizes produced similar execution times, between approximately 0.000278 s and 0.000313 s.

Overall, the best block size depends on the image-processing operation.

